Equation Sheet for MEEN 305 Midterm Exam

Conservation (equilibrium) equations:

$$\sum \vec{F} = 0\ ; \quad \sum \vec{M} = 0$$

$$N_x' = -q_x$$

$$T_x' = -t_x$$

$$M_z'' = V_y' = q_y$$

$$M_y'' = V_z' = -q_z$$

$$P = T\omega \qquad 1\ HP = 550\ \frac{lb_f - ft}{s}$$

Stresses in circular cross-sections:

$$\sigma_{axial} = \frac{N_x}{A}$$

$$\hat{\sigma}_{bending} = \frac{r_o}{I}\sqrt{\left(M_y\right)^2 + \left(M_z\right)^2}$$

$$\hat{\tau}_{torsion} = \frac{T_x\, r_o}{J}$$

$$\hat{\sigma}_{vm} = \sqrt{\left(\frac{|N_x|}{A} + \frac{r_o}{I}\sqrt{\left(M_y\right)^2 + \left(M_z\right)^2}\right)^2 + 3\left(\frac{T_x\, r_o}{J}\right)^2}$$

$$\hat{\sigma}_{vm} = \frac{r_o}{I}\sqrt{\left(M_y\right)^2 + \left(M_z\right)^2 + \frac{3}{4}(T_x)^2} \quad for \quad N_x = 0$$

Stresses in polygonal cross-sections:

For two-fold symmetry with corners at $\left(y = \pm c_y, z = \pm c_z\right)$:

$$\sigma_{axial} = \frac{N_x}{A}$$

$$\hat{\sigma}_{bending} = \frac{M_y c_z}{I_y} + \frac{M_z c_y}{I_z}$$

$$\hat{\tau}_{torsion} = \frac{\eta T_x}{(A_i + A_o)t}$$

$$\hat{\sigma}_{vm} = \sqrt{\left(\frac{|N_x|}{A} + \frac{|M_y| c_z}{I_y} + \frac{|M_z| c_y}{I_z}\right)^2 + 3\left(\frac{\eta T_x}{(A_i + A_o)t}\right)^2}$$

$\eta \approx 1.0 \quad for \quad A_i \gg A\ (always\ the\ case\ in\ this\ class)$
$\eta \sim 4.8 \quad for \quad A_i \ll A$

Failure analysis:

Yielding $FoS = \dfrac{Y}{\sigma_{vm}^*}$

Buckling $FoS = \dfrac{P_{cr}}{|\min(0, N_x)|} = \dfrac{\pi^2 E}{L_e^2}\dfrac{\min\left(I_{yy}, I_{zz}\right)}{|\min(0, N_x)|}$

Superscript * denotes maximum magnitude

Geometric properties in circular cross-section:

$$A = A_o - A_i = \frac{\pi}{4}\left(d_o^2 - d_i^2\right) = \pi\left(r_o^2 - r_i^2\right)$$

$$A_m \equiv \frac{A_o + A_i}{2}$$

$$I = I_y = I_z = I_o - I_i = \frac{\pi}{64}\left(d_o^4 - d_i^4\right) = \frac{\pi}{4}\left(r_o^4 - r_i^4\right)$$

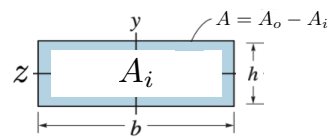
$$J = \iint (y^2 + z^2)\, dA = I_z + I_y$$

Geometric properties in rectangular cross-section:

$$I_z = \frac{1}{12}bh^3 - \frac{1}{12}b_i h_i^3$$

$$I_y = \frac{1}{12}hb^3 - \frac{1}{12}h_i b_i^3$$

$$c_y = \frac{h}{2} \qquad c_z = \frac{b}{2}$$



Rectangular area

Geometric properties in polygonal cross-section:

$$I_z = \sum_{i=1}^{n} I_z^{(i)} + \sum_{i=1}^{n} A_i\left(\Delta y_{cg}^{(i)}\right)^2$$

$$I_y = \sum_{i=1}^{n} I_y^{(i)} + \sum_{i=1}^{n} A_i\left(\Delta z_{cg}^{(i)}\right)^2$$

$$y_{cg} = \frac{1}{A}\sum_{i=1}^{n} A_i\, y_{cg}^{(i)}$$

$$z_{cg} = \frac{1}{A}\sum_{i=1}^{n} A_i\, z_{cg}^{(i)}$$

Deformation constitutive equations for isotropic linear thermoelasticity:

$$\bar{N}_{x,BC} \equiv \frac{1}{L}\int_{x_B}^{x_C} N_x(x)dx \qquad \bar{T}_{x,BC} \equiv \frac{1}{L}\int_{x_B}^{x_C} T_x(x)dx \qquad \bar{M}_{y,BC} \equiv \frac{1}{L}\int_{x_B}^{x_C} M_y(x)dx \qquad \bar{M}_{z,BC} \equiv \frac{1}{L}\int_{x_B}^{x_C} M_z(x)dx$$

$$u' - \alpha\Delta T = \frac{N_x}{EA} \quad \Rightarrow \quad \Delta L_{BC} = \frac{\bar{N}_{x,BC}}{k_{x,BC}} + L_{BC}\alpha_{BC}\Delta\bar{T}_{BC} \qquad k_{x,BC} \equiv \frac{E_{BC}A_{BC}}{L_{BC}}$$

$$\phi' = \frac{T_x}{GJ} \quad \Rightarrow \quad \Delta\phi_{BC} = \frac{\bar{T}_{x,BC}}{\kappa_{x,BC}} \qquad \kappa_{x,BC} \equiv \frac{G_{BC}J_{BC}}{L_{BC}}$$

$$v'' = \theta_z' = \frac{M_z}{EI_z} \quad \Rightarrow \quad \Delta\theta_{z,BC} = \frac{\bar{M}_{z,BC}}{\kappa_{z,BC}} \qquad \kappa_{z,BC} \equiv \frac{E_{BC}I_{z,BC}}{L_{BC}}$$

$$-w'' = \theta_y' = \frac{M_y}{EI_y} \quad \Rightarrow \quad \Delta\theta_{y,BC} = \frac{\bar{M}_{y,BC}}{\kappa_{y,BC}} \qquad \kappa_{y,BC} \equiv \frac{E_{BC}I_{y,BC}}{L_{BC}}$$

Materials	Specific Weight (lb/in³)	Moduls of Elasticity E (10^3) ksi	Modulus of Rigidity G (10^3) ksi	Yield Strength (ksi) σ_Y Tens.	Comp.[b]	Shear	Ultimate Strength (ksi) σ_u Tens.	Comp.[b]	Shear	% Elongation in 2 in. specimen	Poisson's Ratio ν	Coef. of Therm. Expansion α (10^{-6})/°F
Metallic												
Aluminum Wrought Alloys — 2014-T6	0.101	10.6	3.9	60	60	25	68	68	42	10	0.35	12.8
Aluminum Wrought Alloys — 6061-T6	0.098	10.0	3.7	37	37	19	42	42	27	12	0.35	13.1
Cast Iron Alloys — Gray ASTM 20	0.260	10.0	3.9	–	–	–	26	96	–	0.6	0.28	6.70
Cast Iron Alloys — Malleable ASTM A-19	0.263	25.0	9.8	–	–	–	40	83	–	5	0.28	6.60
Copper Alloys — Red Brass C83400	0.316	14.6	5.4	11.4	11.4	–	35	35	–	35	0.35	9.80
Copper Alloys — Bronze C86100	0.319	15.0	5.6	50	50	–	35	35	–	20	0.34	9.60
Magnesium Alloy [Am 1004-T61]	0.066	6.48	2.5	22	22	–	40	40	22	1	0.30	14.3
Steel Alloys — Structural A-36	0.284	29.0	11.0	36	36	–	58	58	–	30	0.32	6.60
Steel Alloys — Structural A992	0.284	29.0	11.0	50	50	–	65	65	–	30	0.32	6.60
Steel Alloys — Stainless 304	0.284	28.0	11.0	30	30	–	75	75	–	40	0.27	9.60
Steel Alloys — Tool L2	0.295	29.0	11.0	102	102	–	116	116	–	22	0.32	6.50
Titanium Alloy [Ti-6Al-4V]	0.160	17.4	6.4	134	134	–	145	145	–	16	0.36	5.20
Nonmetallic												
Concrete — Low Strength	0.086	3.20	–	–	–	1.8	–	–	–	–	0.15	6.0
Concrete — High Strength	0.086	4.20	–	–	–	5.5	–	–	–	–	0.15	6.0
Plastic Reinforced — Kevlar 49	0.0524	19.0	–	–	–	–	104	70	10.2	2.8	0.34	–
Plastic Reinforced — 30% Glass	0.0524	10.5	–	–	–	–	13	19	–	–	0.34	–
Wood Select Structural Grade — Douglas Fir	0.017	1.90	–	–	–	–	0.30[c]	3.78[d]	0.90[d]	–	0.29[e]	–
Wood Select Structural Grade — White Spruce	0.130	1.40	–	–	–	–	0.36[c]	5.18[d]	0.97[d]	–	0.31[e]	–

Materials	Density ρ (Mg/m³)	Moduls of Elasticity E (GPa)	Modulus of Rigidity G (GPa)	Yield Strength (MPa) σ_Y Tens.	Comp.[b]	Shear	Ultimate Strength (MPa) σ_u Tens.	Comp.[b]	Shear	% Elongation in 50 mm specimen	Poisson's Ratio ν	Coef. of Therm. Expansion α (10^{-6})/°C
Metallic												
Aluminum Wrought Alloys — 2014-T6	2.79	73.1	27	414	414	172	469	469	290	10	0.35	23
Aluminum Wrought Alloys — 6061-T6	2.71	68.9	26	255	255	131	290	290	186	12	0.35	24
Cast Iron Alloys — Gray ASTM 20	7.19	67.0	27	–	–	–	179	669	–	0.6	0.28	12
Cast Iron Alloys — Malleable ASTM A-197	7.28	172	68	–	–	–	276	572	–	5	0.28	12
Copper Alloys — Red Brass C83400	8.74	101	37	70.0	70.0	–	241	241	–	35	0.35	18
Copper Alloys — Bronze C86100	8.83	103	38	345	345	–	655	655	–	20	0.34	17
Magnesium Alloy [Am 1004-T61]	1.83	44.7	18	152	152	–	276	276	152	1	0.30	26
Steel Alloys — Structural A-36	7.85	200	75	250	250	–	400	400	–	30	0.32	12
Steel Alloys — Structural A992	7.85	200	75	345	345	–	450	450	–	30	0.32	12
Steel Alloys — Stainless 304	7.86	193	75	207	207	–	517	517	–	40	0.27	17
Steel Alloys — Tool L2	8.16	200	75	703	703	–	800	800	–	22	0.32	12
Titanium Alloy [Ti-6Al-4V]	4.43	120	44	924	924	–	1,000	1,000	–	16	0.36	9.4
Nonmetallic												
Concrete — Low Strength	2.38	22.1	–	–	–	12	–	–	–	–	0.15	11
Concrete — High Strength	2.37	29.0	–	–	–	38	–	–	–	–	0.15	11
Plastic Reinforced — Kevlar 49	1.45	131	–	–	–	–	717	483	20.3	2.8	0.34	–
Plastic Reinforced — 30% Glass	1.45	72.4	–	–	–	–	90	131	–	–	0.34	–
Wood Select Structural Grade — Douglas Fir	0.47	13.1	–	–	–	–	2.1[c]	26[d]	6.2[d]	–	0.29[e]	–
Wood Select Structural Grade — White Spruce	3.60	9.65	–	–	–	–	2.5[c]	36[d]	6.7[d]	–	0.31[e]	–